

Yinchu Li

PhD Candidate · Marine Geophysics

School of Earth and Atmospheric Sciences, Georgia Institute of Technology Atlanta, GA, 30318, USA

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EDUCATION

Georgia Institute of Technology

Atlanta, GA, USA

PhD in Geophysics, GPA: 4.0/4.0

Aug. 2021 -

- Advisor: Dr. Samer Naif

Southern University of Science and Technology

Shenzhen, China

MSc in Mechanics (Geophysics), GPA: 3.76/4.0

Sept. 2018 - June 2020

- Thesis: Deep learning-based electrical imaging monitoring of shale gas fracturing using steel well casings as long electrodes
- Advisor: Dr. Dikun Yang

China University of Petroleum (East China)

Qingdao, China

BEng in Applied Geophysics, GPA: 3.72/4.0

Sept. 2014 - June 2018

- Thesis: Estimation of 2-D site amplification effects: the case of the historical centre of Napoli
- Advisor: Dr. Concettina Nunziata (University of Naples Federico II)

RESEARCH INTERESTS

Controlled-source electromagnetic imaging, Magnetotellurics methods, Geophysical inverse problem, Marine Geophysics, Fluid-tectonic interactions at plate margins, Deep learning for geosciences

RESEARCH EXPERIENCE

Georgia Institute of Technology

Atlanta, GA, USA

Graduate Research Assistant

Spring 2022 - Spring 2024

- Controlled-source electromagnetic (CSEM) and Magnetotellurics (MT) data collected by the NSF funded marine EM survey: *ElectroMagnetic Alaskan GeoPRISMS Experiment* (EMAGE)
- CSEM data processing and navigation.
- Inversion of the CSEM and MT data at the Alaska Subduction Zone jointly and individually.
- Comprehensive interpretation of inversion results from the Shumagin Gap and Semidi segments, Alaska-Aleutian subduction zone.
- Improve CSEM data processing and inversion visualization.

Southern University of Science and Technology

Shenzhen, China

Graduate Research Assistant

Aug. 2020 - June 2021

- Long-electrode electrical monitoring of shale gas fracturing and well casing integrity using surface electrical data and deep learning approaches. The work is a part of the National Natural Science Foundation of China funded research: *High Resolution Imaging Monitoring of Shale Gas Fracturing Fluid using Long Electrode Electromagnetic Method*.
- Collaborated with the BGP company to develop a deep learning-based real-time hydraulic fracturing electrical monitoring system for the field experiment.
- Developed a GUI to customize the deep learning model architecture and parameters with a 3D visualization tool to visualize the results.

- Study Abroad Scholarships
- Finished Bachelor thesis focusing on 2D site amplification effects modeling of the historical centre of Napoli under the supervision of Dr. Concettina Nunziata.

PUBLICATIONS

- Li, Y., S. Naif, K. Key, S. Constable, R. L. Evans, D. J. Shillington, A. Becel, and D. Cordell (2025). “Insufficient fluids to explain weak plate coupling along the shallow megathrust at the Shumagin Gap”. *Nature Communications*. Under review.
- Hu, Y., D. Yang, Y. Li, Z. Wang, and Y. Lu (2022). “3-D Numerical Study on Controlled Source Electromagnetic Monitoring of Hydraulic Fracturing Fluid With the Effect of Steel-Cased Wells”. *IEEE Transactions on Geoscience and Remote Sensing*, 60, pp. 1–10. DOI: 10.1109/TGRS.2021.3100774.
- Li, Y. and D. Yang (2021). “Electrical imaging of hydraulic fracturing fluid using steel-cased wells and a deep learning method”. *Geophysics*, 86, 4, E315–E332. DOI: 10.1190/geo2020-0178.1.

CONFERENCE ABSTRACTS

- Li, Y., S. Naif, D. Cordell, K. Key, R. L. Evans, S. Constable, D. J. Shillington, and A. Becel (2024). “Megathrust heterogeneity revealed by Electromagnetic Data at the Aleutian-Alaska Subduction Zone”. *The 26th EM Induction Workshop 2024*.
- Li, Y., S. Naif, D. Cordell, K. Key, R. L. Evans, S. Constable, D. J. Shillington, and A. Becel (2023). “Electromagnetic Study of Fluid Distribution at the Aleutian-Alaska Subduction Zone”. AGU23.
- Wang, Z., Y. Lu, Y. Hu, Y. Li, K. Wang, and D. Yang (2022). “3D real-time imaging for electromagnetic fracturing monitoring based on deep learning”. *Second International Meeting for Applied Geoscience & Energy*, pp. 702–706. DOI: 10.1190/image2022-3737841.1.
- Naif, S., Y. Li, K. Key, R. Evans, S. Constable, and D. Cordell (2021). “Preliminary results from the Electromagnetic Alaskan GeoPRISMS Experiment (E-MAGE)”. *AGU Fall Meeting Abstracts*. Vol. 2021, GP25A-0402.
- Li, Y. and D. Yang (2020a). “Imaging of steel casing's conductivity using surface electrical data and a deep learning approach”. *SEG Technical Program Expanded Abstracts 2020*, pp. 636–640. DOI: 10.1190/segam2020-3426575.1.
- Li, Y. and D. Yang (2020b). “Monitoring the integrity of steel well casings using electrical data on the surface”. *SEG 2020 Workshop: 2nd SEG Borehole Geophysics Workshop*, pp. 152–156. DOI: 10.1190/bhgp2020-41.1.
- Li, Y. and D. Yang (2019). “Fast electrical imaging of injected fluid in hydraulic fracturing using a practical interactive parameter estimation method”. *SEG Technical Program Expanded Abstracts 2019*, pp. 1024–1028. DOI: 10.1190/segam2019-3215758.1.

- Yang, D. and Y. Li (2019). “Monitoring directional fluid flow in shale gas hydraulic fracturing through electrically energized steel well casings”. *International Workshop on Gravity, Electrical & Magnetic Methods and Their Applications, Xi'an, China, 19-22 May 2019*, pp. 45–48. DOI: 10.1190/GEM2019-012.1.
- Yang, D., Y. Li, X. Liu, F. Zhu, and J. Zhang (2019). “Feasibility Study of Electrical Monitoring of In-situ Thermal Desorption in Remediation of Contaminated Soil”. *AGU Fall Meeting Abstracts*. Vol. 2019, NS21C-0831.

HONORS & AWARDS

2025 **2025 School of Earth and Atmospheric Sciences Departmental Travel Fund**
 2025 **2025 NSF GAGE/SAGE Community Science Workshop Travel Scholarship**
 2024 **26th EM Induction Workshop Financial Support**, 26th EM Induction Workshop
 2021 **Best Presentation**, The 15th China International Geo-electromagnetism Workshop
 2020 **Best Student Presentation**, 2nd SEG Borehole Geophysics Workshop
 2020 **SEG Travel Grant (virtual registration grant)**, SEG International Exposition and 90th Annual Meeting
 2019 **Outstanding Student Paper**, Annual Meeting of Chinese Geoscience Union
 2019 **SEG Travel Grant (Qualified before VISA)**, SEG International Exposition and 89th Annual Meeting
 2018 **Outstanding graduate**, China University of Petroleum (East China)

FIELD EXPERIENCE

Near Trench Community Geodetic Experiment

Alaska-Aleutian Trench

Graduate Student

June/July 2024

- Participation in 13-days seafloor geodetic instruments deployment at sea aboard the R/V Sikuliaq under *Near-Trench Community Geodetic Experiment*.
- Deployed multiple GNSS-Acoustic benchmarks and transponders for measuring deformation and coupling along the Alaska-Aleutian megathrust. Two wavegliders were deployed for GNSS-A surveys and conductivity, temperature, and depth (CTD) data was collected for a sound speed profile.
- PIs on board: Dr. Andrew Newman, Georgia Institute of Technology, Dr. Spahr Webb, Lamont-Doherty Earth Observatory and Dr. Surui Xie, University of Houston.

Kivu Project

Western Rwanda

Graduate Student

May/June 2023

- Participation in 22-days land Magnetotellurics field work under an NSF funded collaborative research: *Constraining transient magma intrusion processes in the Nyiragongo-Kivu continental rift zone*.
- PIs: Dr. Andrew Newman and Dr. Samer Naif, Georgia Institute of Technology.

Seismology field work

Summerville, SC, USA

Graduate Student

May 2023

- Participation in 4-days nodal seismic deployment for understanding seismicity distribution within the South Eastern United States.
- PI: Dr. Zhigang Peng, Georgia Institute of Technology.

Oceanographic field study

Savannah, GA, USA

Graduate Student

May 2023

- Participation in Ocean Science and Engineering (OSE) retreat and collaborative communication with the Skidaway Institute of Oceanography (SkIO), University of Georgia.
- One day on board the R/V Savannah to learn the basic practical knowledge of oceanography.

Seismology class field trip

Columbia, SC, USA

Graduate Student

Oct. 2022

- Participation in 4-days nodal seismic deployment near the town of Elgin to study the tremors
- EAS 6314/4314 Seismology (Fall Semester 2022) given by Dr. Zhigang Peng, Georgia Institute of Technology.

Marine electromagnetic survey HT-RESIST

Hikurangi Trench

Graduate Student

Feb. 2019

- Participation in 2nd cruise of the *Hikurangi Trench Regional Electromagnetic Survey to Image the Subduction Thrust* (HT-RESIST) aboard R/V Roger Revelle.
- Ocean bottom electromagnetic stations recovery.
- PIs: Dr. Samer Naif and Dr. Kerry Key, Columbia University, Lamont-Doherty Earth Observatory.

Long-electrode electrical monitoring of shale gas fracturing

Weiyuan, China

Graduate Student

May/June 2021

- Software development and test for the long-electrode electrical monitoring method.
- Real-time hydraulic fracturing electrical monitoring data collection, collaborating with BGP.

Long-electrode electrical method validation experiment

Huaibei, China

Graduate Student

Apr. 2021

- 3-days field work for collecting long-electrode electrical data with collaborators from China University of Mining and Technology.

Field work

Eastern Tibetan Plateau, China

Graduate Student

Sept. 2020

- Participation in scouting part of the MT field work: India-Asia collision process through a study of Gongga Shan.

Field trip

Xinwen, China

Undergraduate Student

June 2015

- 12 days field trip for undergraduate course: Fundamentals of Geology.

TEACHING EXPERIENCE

Georgia Institute of Technology

Atlanta, GA, USA

Teaching Assistant

Fall 2021, Fall 2024, Summer 2025

- EAS 2600: Earth processes, Lab session
- EAS 1600: Introduction to Environmental Science, Lab session
- Participate in and help design/update some of the lab assignments.
- Gave a guest lecture in a EAS 1600 class

Southern University of Science and Technology

Shenzhen, China

Teaching Assistant

Feb. - June 2019

- ESS 302: Applied Geophysics II (Gravity, Magnetic, Electrical, Electromagnetic and Well Logging).

PRESENTATIONS

- **Li, Y., S. Naif, D. Cordell, K. Key, R. L. Evans, S. Constable, D. J. Shillington, and A. Becel (2024).** "Megathrust heterogeneity revealed by Electromagnetic Data at the Aleutian-Alaska Subduction Zone". The 26th EM Induction Workshop 2024, Beppu, Japan (Oral)
- **Li, Y., S. Naif, D. Cordell, K. Key, R. L. Evans, S. Constable, D. J. Shillington, and A. Becel (2023).** "Electromagnetic Study of Fluid Distribution at the Aleutian-Alaska Subduction Zone". AGU Annual Meeting 2023, San Francisco, CA (Oral)

- Naif, S., **Y. Li**, K. Key, R. Evans, S. Constable, and D. Cordell (2021). “Preliminary results from the Electromagnetic Alaskan GeoPRISMS Experiment (E-MAGE)”. AGU Fall Meeting 2021, New Orleans, LA (Poster)
- **Li, Y.** and D. Yang (2020), Monitoring the integrity of steel well casings using electrical data on the surface”. SEG 2020 Work- shop: 2nd SEG Borehole Geophysics Workshop, Virtual (Poster)
- **Li, Y.** and D. Yang (2020). “Imaging of steel casing’s conductivity using surface electrical data and a deep learning approach”. SEG International Exposition and 90th Annual Meeting, Virtual (Poster)
- **Li, Y.** and D. Yang (2019). “Fast electrical imaging of injected fluid in hydraulic fracturing using a practical inter- active parameter estimation method”. SEG International Exposition and 89th Annual Meeting, San Antonio, TX (Oral)

SKILLS

Programming languages	MATLAB, Python, Typescript, C, Fortran
Visualization	GMT, ParaView, PyQt/PySide, PyVista, Adobe Illustrator, Geomap App
Geophysical inversion code	MARE2DEM, OCCAM1DCSEM
Machine learning frameworks	TensorFlow, Keras, Scikit-learn
Languages	Chinese (Native), English (Professional proficiency)

PROFESSIONAL SERVICES

PEER-REVIEWER

GEOPHYSICS, Journal of Applied Geophysics

AFFILIATIONS

- American Geophysical Union
- Society of Exploration Geophysicists